

36. Let $S: \mathbb{R}^p \rightarrow \mathbb{R}^n$ and $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear transformations. Show that the mapping $x \mapsto T(S(x))$ is a linear transformation (from $\mathbb{R}^p$ to $\mathbb{R}^m$). [Hint: Compute $T(S(cu+dv))$ for u, v in $\mathbb{R}^p$ and scalars c and d . Justify each step of the computation, and explain why this computation gives the desired conclusion.]

Solution:

$$\begin{aligned}
 T(S(cu+dv)) &= T(S(cu) + S(dv)) \quad (\text{Condition 1}) \\
 &= T(cS(u) + dS(v)) \quad (\text{Condition 2}) \\
 &= T(cS(u)) + T(dS(v)) \quad (\text{Condition 1}) \\
 &= [cT(S(u)) + dT(S(v))] \quad (\text{Condition 2})
 \end{aligned}$$

c and d are arbitrary scalars. Let $c=1$ and $d=1$

$$\therefore T(S(u+v)) = T(S(u)) + T(S(v)) \rightarrow \text{Condition 1 satisfied.}$$

$$\text{Let } d=0. \therefore T(S(cu+0v)) = T(S(cu)) = cT(S(u)) + 0 \cdot T(S(v)) = cT(S(u))$$

$\therefore$ Condition 2 satisfied. $\therefore x \mapsto T(S(x))$ is a linear transformation.